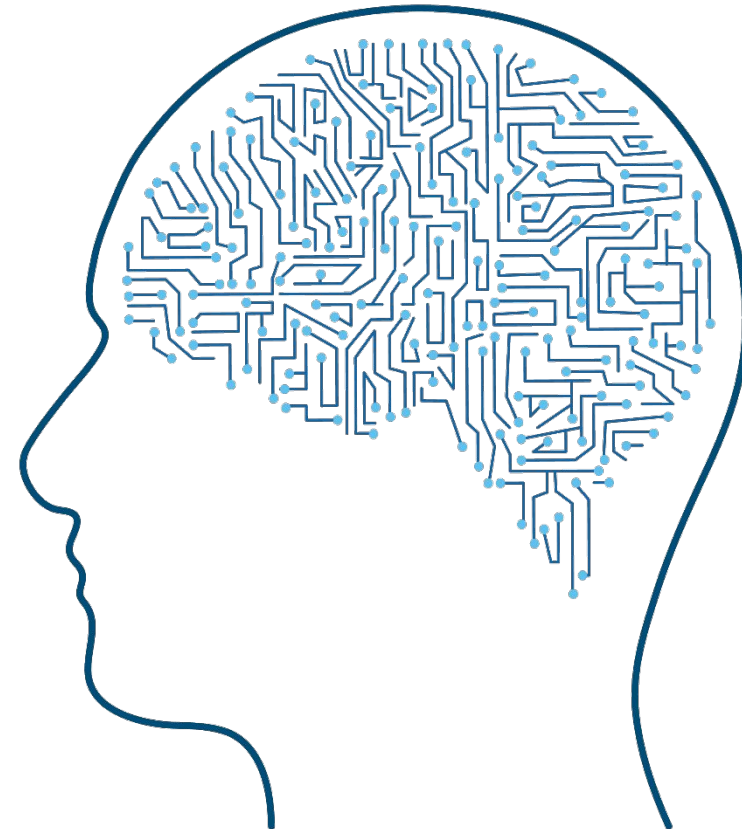


Ubiquitous Computing

Exercise Session

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Exercise:

Steps in designing a study

1. Consider your **independent** and **dependent variables** and how they are related
2. Design experimental treatments to manipulate your independent variable
3. Write a specific, testable **hypothesis**
→ reformulate as a **null hypothesis**
4. Assign subjects to groups, either **between-subjects** or **within-subjects**
5. Plan how you will **measure** your dependent variable, and what statistical analysis on your data makes sense

0. Use one of these examples, or come up with your own:

The more smartphone use in the evening, the more restless your sleep.

Students that complete assignments with pen and paper can remember the material better the next day than those who type on a computer.

Daily use of smart earbuds leads to users getting worse in blending out background noises.

Typing longer messages on a smartphone's virtual keyboard is much faster than dictating.

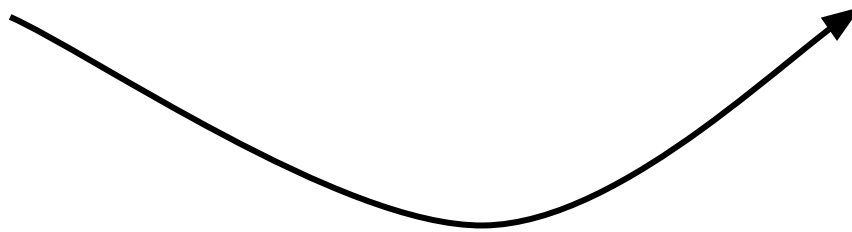
People that enter a PIN code through a smart watch App make more mistakes than on an ATM console.

1. Identify the independent and dependent variables, and how they relate

People drinking coffee in the morning tend to walk more during the day.

cups of coffee
(0, 1, 2, ...)

steps taken over the entire
day



more morning cups of coffee leads to more
steps during entire day afterwards

2. Design experimental treatments to manipulate your independent variable

People drinking coffee in the morning tend to walk more during the day.


cups of coffee
(0, 1, 2, ...)

Some participants are asked to drink 2 cups of coffee in the morning.

Some participants are asked to not drink coffee at all in the morning.

3. Write a specific, testable hypothesis, and reformulate as a null hypothesis

Study participants that drink 2 coffees in the morning will have higher step counts than those participants that don't drink coffee at all in the morning.

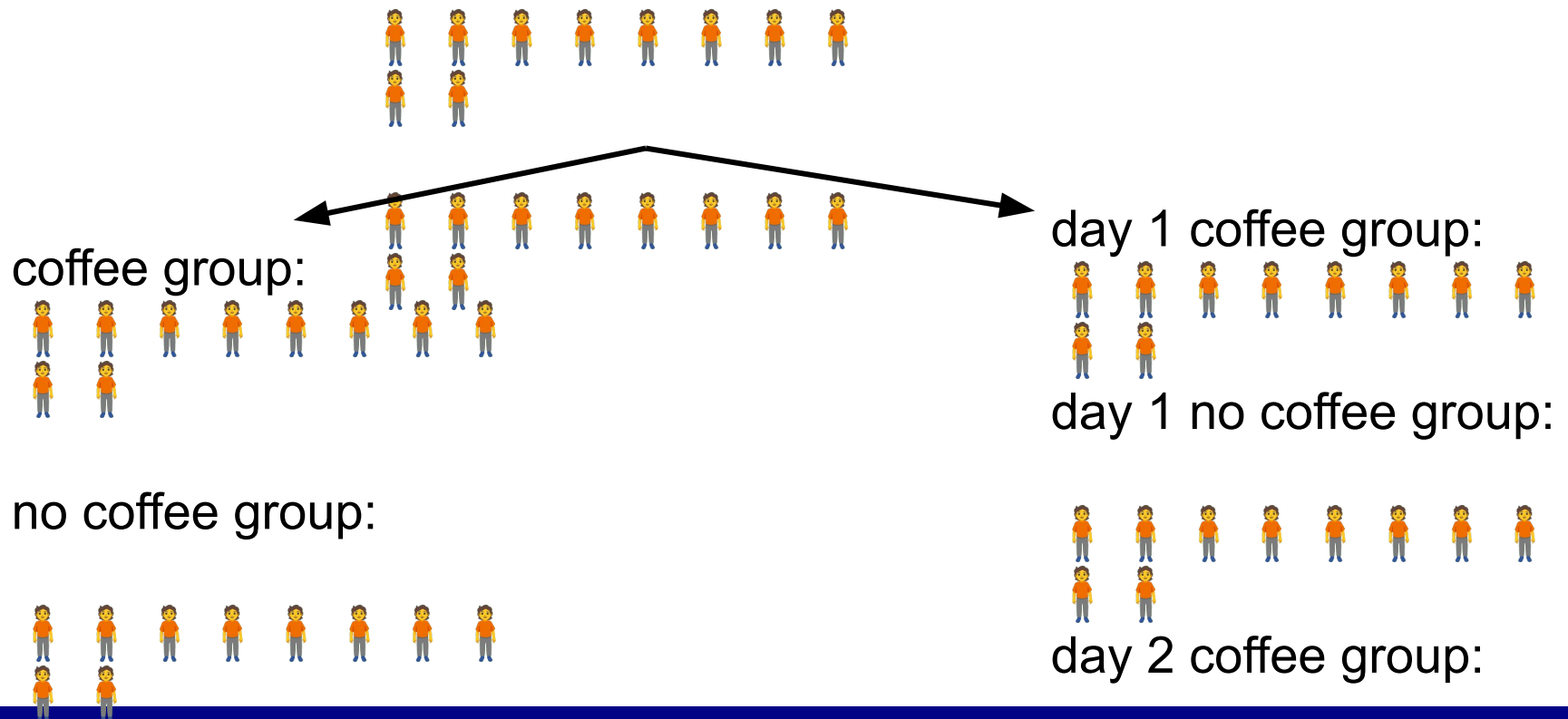
Null Hypothesis:

The difference in steps walked by participants that drink coffee and the steps walked by those that didn't drink coffee in the morning *is up to randomness*.

4. Assign subjects to groups, either between-subjects or within-subjects

between-subjects:

within-subjects:



5. Plan how you will measure your dependent variable, and what statistical analysis on your data makes sense

People drinking coffee in the morning tend to walk more during the day.
steps taken over the entire day

All participants will wear a step counter for the entire day after breakfast, till they go to bed, producing a single step count per participant, per day.

What could introduce bias?

The length of the day, the person's routines or the weekday, ...

5. Plan how you will measure your dependent variable, and what statistical analysis on your data makes sense

All participants will wear a step counter for the entire day after breakfast, till they go to bed, producing a single step count per participant, per day.

no coffee group		coffee group	
participant 1	6293	participant 2	6343
participant 4	7232	participant 3	7329
participant 7	8126	participant 5	7129
participant 8	4282	participant 9	6278
...	...	...	...

mean: 6070

mean: 7802

**+ significance test
(e.g., $p < 0.05$)**

5. Plan how you will measure your dependent variable, and what statistical analysis on your data makes sense

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